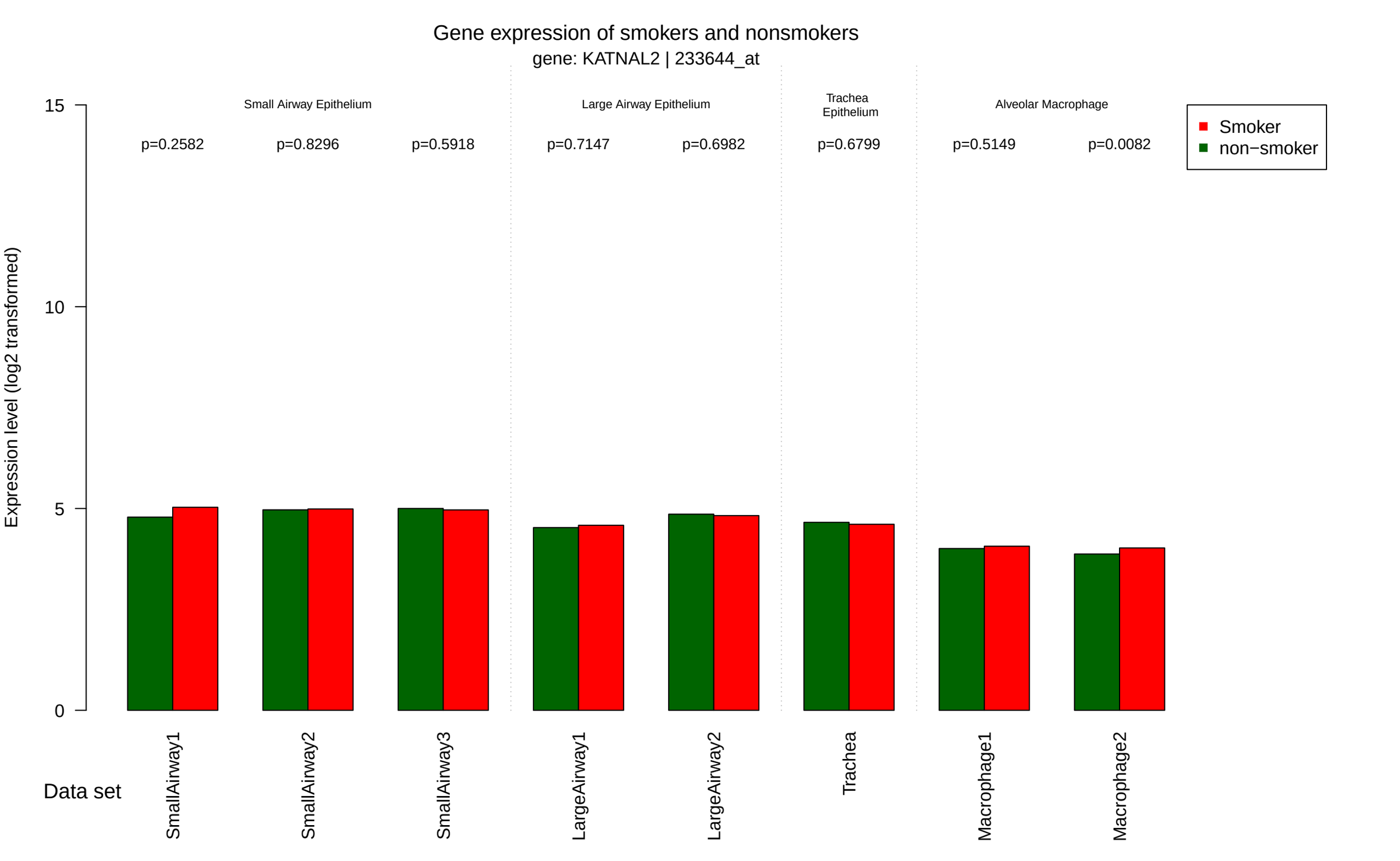
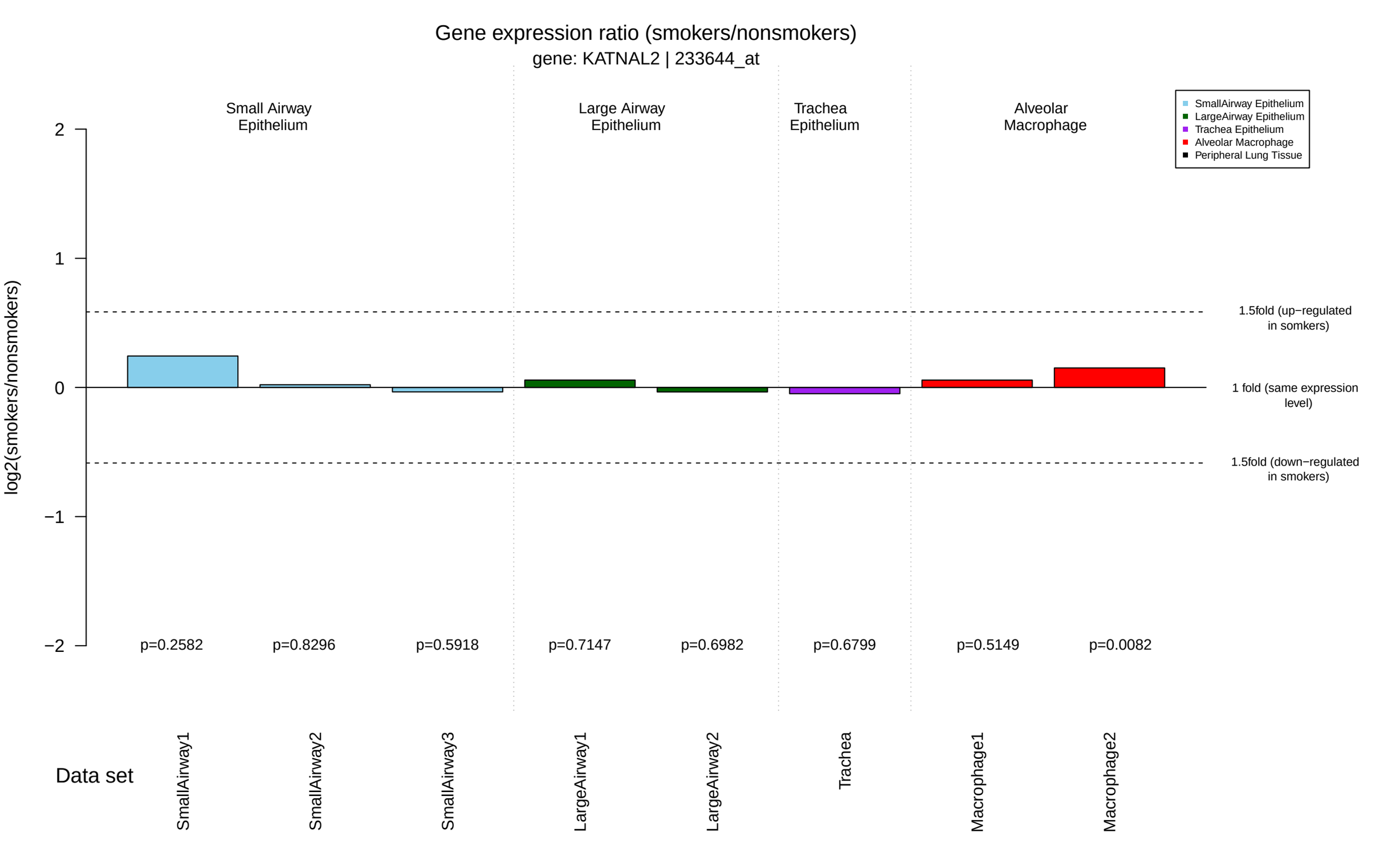
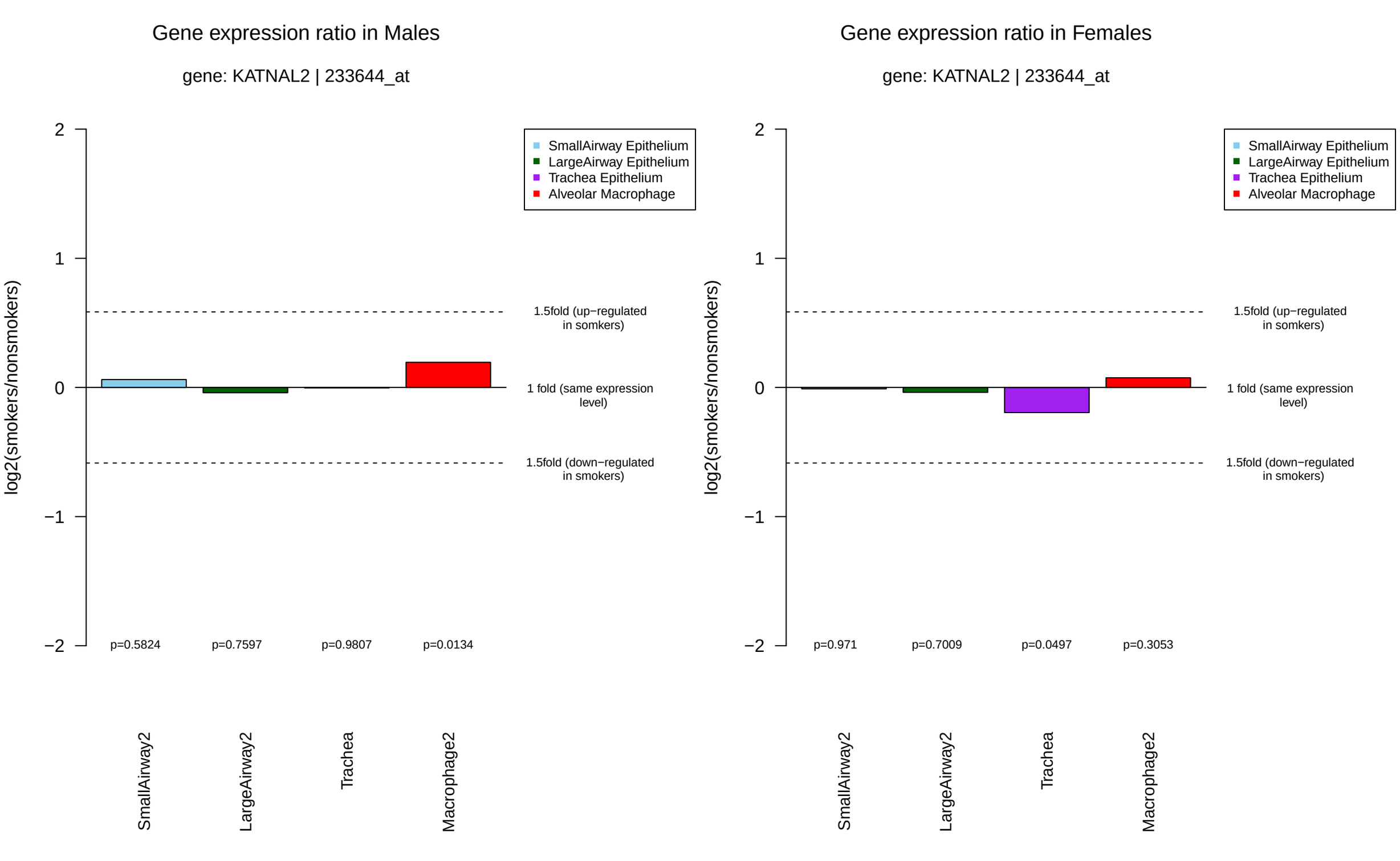
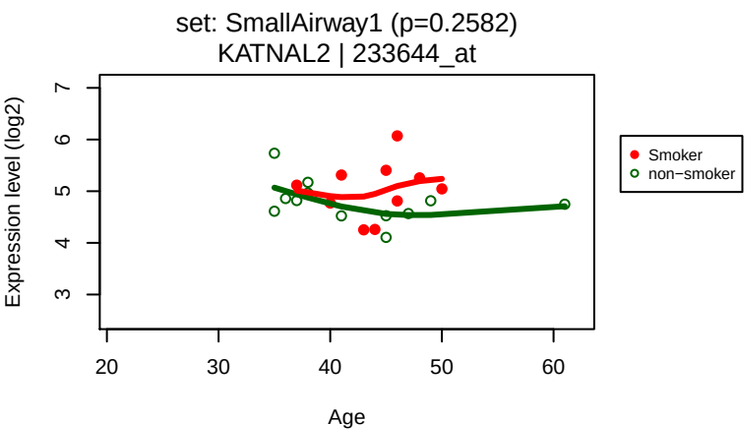
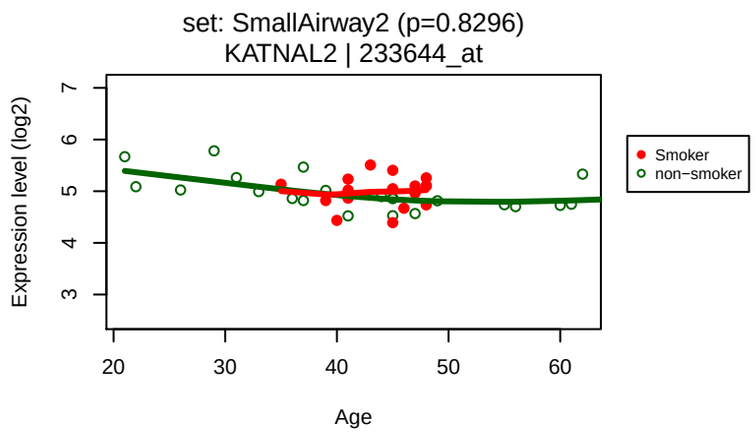
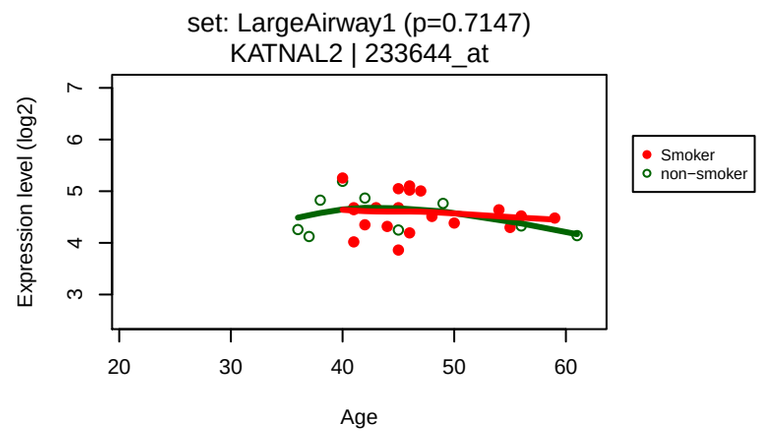

Gene expression of smokers and nonsmokers
gene: KATNAL2 | 233644_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.2582
p=0.8296
p=0.5918
p=0.7147
p=0.6982
p=0.6799
p=0.5149
p=0.0082
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

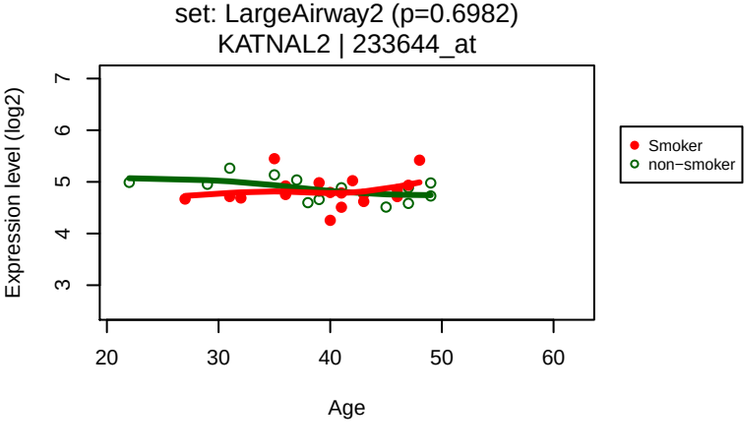

Gene expression ratio (smokers/nonsmokers)
gene: KATNAL2 | 233644_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2582
p=0.8296
p=0.5918
p=0.7147
p=0.6982
p=0.6799
p=0.5149
p=0.0082
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

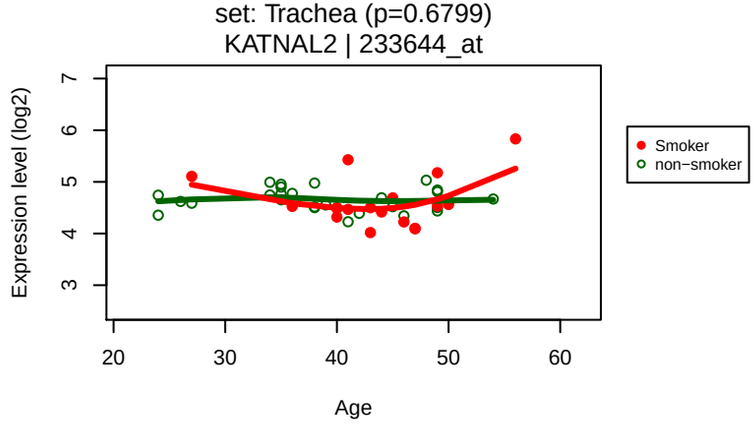

Gene expression ratio in Males
gene: KATNAL2 | 233644_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5824
p=0.7597
p=0.9807
p=0.0134
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: KATNAL2 | 233644_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.971
p=0.7009
p=0.0497
p=0.3053
SmallAirway2
LargeAirway2
Trachea
Macrophage2

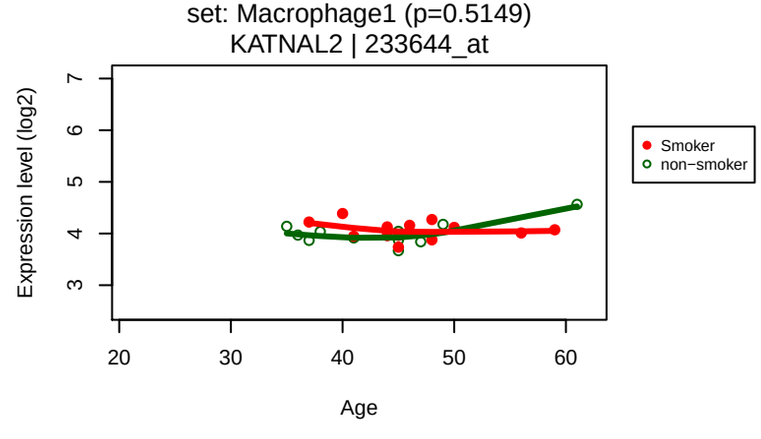

set: SmallAirway1 (p=0.2582)
KATNAL2 | 233644_at
Expression level (log2)
Age
Smoker
non−smoker

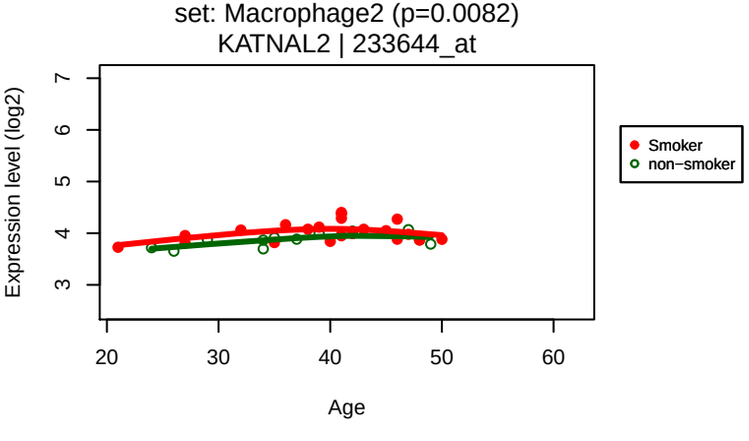

set: SmallAirway2 (p=0.8296)
KATNAL2 | 233644_at
Expression level (log2)
Age
Smoker
non−smoker

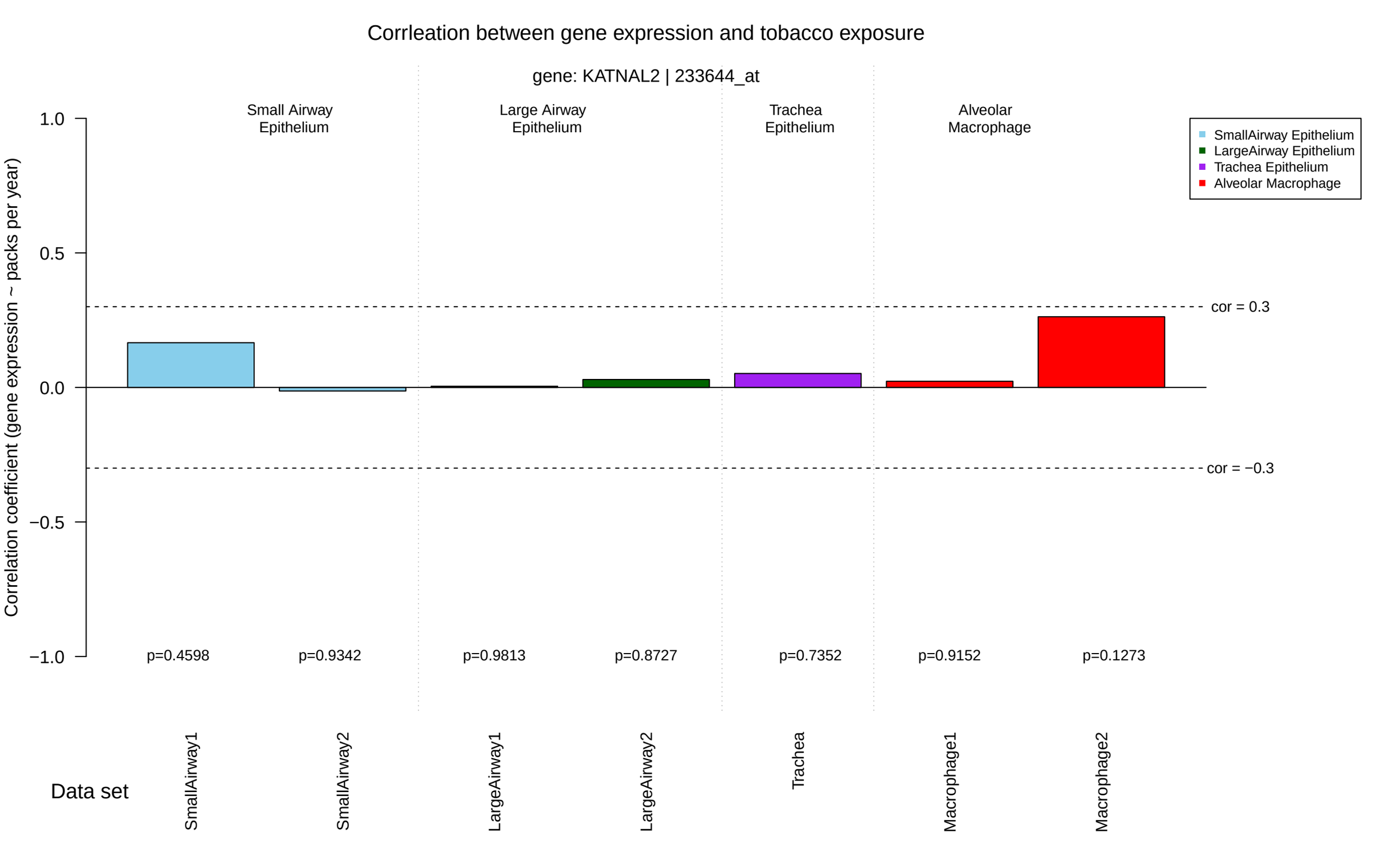

set: LargeAirway1 (p=0.7147)
KATNAL2 | 233644_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.6982)
KATNAL2 | 233644_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.6799)
KATNAL2 | 233644_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.5149)
KATNAL2 | 233644_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0082)
KATNAL2 | 233644_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: KATNAL2 | 233644_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
Correlation coefficient (gene expression ~ packs per year)
p=0.4598
p=0.9342
p=0.9813
p=0.8727
p=0.7352
p=0.9152
p=0.1273
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2